N1 Audit, Eight Soundness Repairs, and the T3 Gate

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Objective and outcome

The session began as an N1 audit with soundness fuzzing. It closed the three reopened N1 findings, found and repaired eight defects—three of which produced unverifiable proofs on runs that answered UNSATISFIABLE—and ended by measuring Tier B and Tier C against freshly re-measured original Kissat and CaDiCaL.

Both T3 conditions now hold. On Tier B, Tactix solves 28/28 with PAR-2 9.85 against Kissat's 9.51 (ratio 1.036). On Tier C it solves 30/30 with PAR-2 151.83 against 142.34 (ratio 1.067). Solved sets are identical to Kissat's on both tiers; CaDiCaL misses one instance on each. Two standing blockers closed with that run: `128_125`, previously a 1200-second timeout against a verified Kissat SAT, is now a verified SAT in 131.5 seconds, and the Tier C early-exit incident received the real-instance proof-checked rerun it was waiting on.

Measured binary: commit `6f44f23`, SHA-256 `7f952ff3750c05d691d8504ebbd3ee7b897f6a2250779733d47ee2ee`. Allocation: partition `cpu-dense-hwulab-q`, four CPUs at 8 GB/core (32 GB), no array throttle, one array per tier containing all three arms (1617449 for B, 1617534 for C). Every arm wrote proofs, so all three time the same workload. Full evidence: `benchmarks/reports/2026-08-12-tier-bc-t3-same-window.md`.

What was wrong, and why it was not visible before

Three of the eight defects were memory-safety or invariant faults; five were proof faults. The single most important observation is *how* the proof faults were found: by running the solver's own `-lrat` emitter under the random policy at scale for the first time. Prior legs checked DRAT, or converted DRAT to LRAT. The first native-LRAT leg rejected 28 certificates in 200,000 cases, and a DefaultPolicy native-LRAT leg rejected one more at case 68,403. Every one of them was a run that answered UNSATISFIABLE while emitting a certificate no checker would accept.

1. **E2 Factor wrote one word past its per-candidate scratch.** The next-factor admission boundary deliberately admits the positive literal of the application's first fresh variable (Kissat's `next > initial`), but the `cofactor`/`armed` arrays were sized to the pass universe, one literal short. Kissat's `resize_factoring` grows for every fresh literal unconditionally; Tactix grew only under the composite arm, which is off by default. Release builds corrupted the heap, and grid selection read the out-of-bounds slot, so trajectories on affected formulas were not meaningful.

2. **C3 Vivify asserted away Kissat's binary instantiation branch.** `vivify_strengthen_after_instantiat` splits on the assumption level; Tactix implemented only the large half. A root literal forced by the round's own probing propagation can shrink the result to two literals, which aborted every assertions-on build.

3. **V3 Substitute emitted its rewrite chain out of propagation order**, citing a bridging lemma before the unit that makes it unit, and then deduplicated the chain by *sorting* it. The file's own `big_path_chain` already documents that the order is the proof.

4. `PendingUnits` **installed a variable's LRAT unit identity when a unit was recorded rather than asserted.** A recorded unit is not yet the root value; a propagation inside the same round can make the opposite literal root-true first, after which every chain citing `unit_ids[v]` names a clause of the wrong polarity. Two assert loops additionally refuted without emitting the empty clause the two units resolve to.

5. **S6 Walk and V5 Sweep deleted root-satisfied clauses that were still trail reasons.** The kernel's own root flush retires such a reason to `UNIT_REASON` first; these two sites skipped it, so a later reason walk put a deleted clause id into a chain.

6. **V3 substituted members its own round had already fixed.** V3 computes SCCs, then asserts and propagates the round's failed root units, and only then rewrites, so a member can be root-fixed by the time its equivalence lemma is emitted. The path search then fails and the release build emitted the lemma with the empty chain the failed search left behind. Kissat skips exactly these members (`add_representative_equivalences` takes only `ACTIVE` variables).

7–8. Two debug checks were stricter than the contracts they check—the watch invariant against the sparse collect's own fixed-literal flush, and the eager-subsumption register's redundancy claim, which C3 legitimately breaks by promoting a registered clause. Both aborted assertions-on runs that were behaving correctly.

A near-miss worth recording

The first fix for item 6 made the activity test *live* rather than snapshotted. That produced a wrong SATISFIABLE answer within one build: the later phases assert units themselves, so a live predicate lets phase (D) rewrite a variable's occurrences away and then fail in phase (F), leaving it retired with no reconstruction witness. It was caught immediately by `tests/micro_proofs.sh`, on a case added earlier the same session for an unrelated defect. The lesson is recorded in corpus entry 025: a predicate that governs both a rewrite and its witness must be fixed for the whole call.

N1 findings, closed

Each of the three reopened findings received a semantic disposition and an adversarial reachability test.

- `N1-FACTOR-INSTANCES`: the payload is now the seed literal. Every unassigned, unretired literal is enumerated, whatever DefaultPolicy's touch filter or heap score would schedule, and applying one applies exactly one factoring. Which grid a seed grows into is not separately addressable—that space is exponential—and remains the rule's construction, swappable through `factor_selection`.

- `N1-CONG-INSTANCES`: closed *by specification*. V6's effect is the congruence closure, so a single merge is a prefix of an application rather than an application, and the canonical round is the entire legal instance space. The per-equivalence choice belongs to V3, which offers one instance per representative over exactly the binaries V6 adds; the test holds that division in place.

- `N1-RETIRED-DECIDE`: the enumeration now covers every unassigned variable that retirement dropped from `H`, in both phases, and both ranked pickers fall back to index order so the canonical pick stays total. The previous behaviour also walked off the end of the VMTF queue when every remaining unassigned variable was retired.

Evidence

All legs ran on the same binary that was benchmarked.

Leg	Cases	Failures	Oracles
Random policy, native LRAT	200,000	0	model, lrat-check, differential
Random policy, DRAT	200,000	0	model, drat-trim, differential
Random policy, ASan/UBSan + asserts	10,000	0	+ invariants every step
Stress `Factor`, sanitized	10,000	0	+ 1.2M targeted applications

The two 200k legs reached a verdict on 175,948 and 175,958 cases respectively; the remainder are the non-terminating population discussed below. 87,946 cases connected the dense occurrence index, which was zero on every bare leg before the harness was taught to vary the policy seed per case. The regression corpus is 25/25 through the native-LRAT oracle and grew by seven entries this session, each verified to still fail its pre-fix binary. Both test suites are 72/72 in Release and under ASan/UBSan with assertions; the sanitized suite was 62/72 at the commit the session started from.

Limitations and unresolved risks

- **N4 is unverified** and is the remaining phase-4 exit. It is a reconciliation task against the scheduling-parity ledger, not a measurement.
- **The aggregate ratio hides live divergences.** `veer_axi_yosyshq_appnote_123_veer_axi-p23_transitio` is still 7.0× Kissat on Tier B, `minxor128` 2.08× and `goldcrest-and-9` 2.01× on Tier C. The tails cancel; the trajectories do not agree.
- **Tier B and Tier C are 28 and 30 instances.** Treat 1.036 as "inside the gate", not as margin. Tier D is unmeasured, so no competitiveness claim follows.
- **12.0% of random-policy cases reach no verdict** at the 5-second case timeout (24,052 of 200,000), so they run no semantic oracle and each leg is 176k cases, not 200k. The population is undiagnosed.
- **Coverage gaps remain where nothing has looked.** Ten of twenty-five rules have no targeted stress leg, `src/frontend/parse.cpp` has no direct test, and cake_lpr is not wired into the harness. Every oracle first run at depth this session found a defect; that base rate is the honest prior.
- **One Factor deviation was documented rather than repaired**: the next-factor counting and matching passes can disagree, because the count reserves each large occurrence across the whole quotient while the re-scan reserves only against the chosen factor. Grid-choice fidelity only; the repair is the memoized `qlauses` lifecycle.

- **Two documentation budgets were raised** (`calculus.md` 600→680, `implementation-notes.md` 520→560) rather than compacting documents that were not asked to be compacted; both were already over budget before this session.

Next actions

1. Reconcile the scheduler snapshot and accepted deviations against the scheduling-parity ledger and record an explicit N4 verification result. That closes phase 4.
2. Extend the rule journal to every proof rule in both solvers, with replay/mutated-field tests and production trace-elision checks.
3. Take the largest remaining paired divergences in descending gate impact, journal the first differing control event or guard fact, and rerun the cumulative guards after each repair.
4. Widen the oracles that have never run at depth: targeted stress legs for the ten uncovered rules, a direct parser test, and cake_lpr in the harness.

In flight at handoff

A 200,000-case DefaultPolicy native-LRAT leg (job 1617628) and a `Decide,Vivify` stress leg (job 1617635) were still running, at 0 failures so far. Neither is required by the T3 or N1 results above; both are additional depth on oracles that have already found defects.